Pankaj

Technical Report

on

Geotechnical Investigation

for

Proposed Mono Pole

SITE ID - IN- 3154999

SITE NAME.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State Gujarat, Picode 385310, Police station

SITE ADD.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State Gujarat, Picode 385310, Police station Samarvada

Name of Client

INDUS TOWER LIMITED

Name of Consultant

M/s Multi Project Consulting Engineering

VISHWAS GEOTECH
PRIVATE LIMITED

VISHWAS GEOTECH PVT LTD.

ISO - 9001 CERTIFIED LABORATORY

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Ref. No.: VGPL/ 22-23 / M-0503 **Date : 28/11/2022**

To,
INDUS TOWER LIMITED

Sub. : Soil Investigation work for the Proposed

Proposed Mono Pole
i Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State

SITE ADD.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State Gujarat, Picode 385310, Police station Samarvada

Dear Sir,

The soil investigation report is presented herewith alongwith the properties of sub soil investigation, analysis of strength characteristics, conclusion & recommendation for designing the foundation of proposed structure.

We are enclosing one copies of Test Report.

Thanking you, we remain,

Yours truly,

For **Vishwas Geotech Pvt.Ltd.**

Sign

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GEOTECHNICAL INVESTIGATION REPORT

1 Introduction:

For foundation analysis of the structure on the site, it is necessary

1. To determine soil profile of the site.
2. To know physical properties and strength characteristics of soil at various depths.
3. To determine the Pile Load Capacity of the soil at different depths for the different dia of footings for the test point.
4. To locate the presence of ground water level.
5. To give conclusion and recommendation.

The soil exploration work was carried out at the project site / location given by the client.

The following points were decided:

1. Bore hole - 1
2. Depth of Borehole **10.45** meters

Based on the above points the detailed Geotechnical Program includes

A Field Investigation

1. Drilling of Bore hole
2. Collection of soil samples (Disturbed & Undisturbed)
3. Conducting standard penetration test

B Laboratory Investigation

1. Bulk density and moisture contents
2. Grain size analysis
3. Index properties
4. Shear tests

C Recommendations

Based on the above investigations, the results are to be obtained. The findings would be based on interpretation of results, analysis and computations as per relevant Indian standards.

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2 Field Investigation

2.1 *Boring*

Bore Hole No.	*Location*	*Depth Investigation (m)*
1	*SITE NAME.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje* *SITE ADD.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State Gujarat, Picode 385310, Police station Samarvada*	*10.45*

The exploratory boreholes of 100/150 mm diameter were drilled by Machine Drilling method / Handauger Drilling method The depth of the test bore at the proposed location is **10.45** Meter

2.2 *Sampling*

2.2.1 *Disturbed samples*

Disturbed samples were collected during boring and also from the split spoon sampler. The samples collected were logged, labeled and placed in polyethylene bags and sent to laboratory for testing.

2.2.2 *Undisturbed samples*

Undisturbed samples were collected in thin walled Shelby tubes, sealed with wax, labeled and transported to laboratory for testing.

2.2.3 *Standard Penetration Test*

The standard penetration tests were conducted in accordance with IS 2131-1981. The test gives N – Value, which is the number of blows of last 30cms penetration of split spoon sampler with 65Kg. hammer free fall from 75cms height.

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3 Laboratory Investigation

The following laboratory tests were conducted on undisturbed and disturbed soil samples collected from various depths to find physical properties and strength characteristics. To measure the soil properties in the laboratory, following tests were conducted.

Sr. No.	Tests	Recommended Procedures	Type Samples
1	Sample Preparation	IS 2720 Pt. I	DS / UDS
2	Moisture Contents	IS 2720 Pt. II	DS / UDS
3	Dry Unit Weight	LAMBE	UDS
4	Specific Gravity	IS 2720 Pt. III	DS
5	Liquid Limit	IS 2720	DS
6	Plastic Limit	IS 2720 Pt. V	DS
7	Grain Size Analysis	IS 2720 Pt. IV	DS
8	Soil Classification	IS 1498	DS / UDS
9	Direct Shear Test	IS 2720 Pt. XIII	UDS

4 Results

1 P.L.C.C. are shown in Summary.

2 The laboratory test results of borehole are shown in Table No. 1.

3 The Borelog details of borehole are shown in Table No.-2

4 The Location plan of borehole is shown in Figure No.1

5 Soil Stratification

Borehole No.	Depth (meter)	Description
BH - 01	0.00 to 10.45	Non Plastic Silty Sand

5B Water Chemical

Type of Test	Units	Depth (meter)	Remark
		NA	
PH		NA	
Cholride Content	in ppm	NA	
Sulphate Content (So3)	in ppm	NA	

5C Soil Chemical

Type of Test	Units	Depth (meter)	Remark
		3.0 mtr	
PH		7.2	
Cholride Content	in ppm	60	
Sulphate Content (So3)	in ppm	15	

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6 Typical Calculation for Net Safe Bearing Capacity of soil based on field & laboratory test results:

(A) Typical Calculation for Net Safe Bearing Capacity (As per IS 6403 – 1981) based on Shear Parameter Criteria:

On the basis of available structural details and load on foundation, following assumptions/ adoptions are made for the bearing capacity. Computation for Bore No. 1 at **3.00** m below ground level as per I.S. : 6403 - 1981.

1.Width of foundation	=	4.000	m footing		
2.Depth of foundation	=	3.000	m.		
3.Cohesion ©	=	0.000	kg/cm^2	=	0.000 t/m^2
4.Field Dry density (FDD)	=	1.600	gm/cc	=	1.600 t/m^3
5.Field Moisture Content (FMC)	=	2.500	%		
6.Specific Gravity	=	2.640			
7.Angle of internal friction (∅)	=	30.00	^{o}		
8.Soil Strata	=	--			

Typical calculation:

(1) Size of footing:

Width of footing (B) = 4.000 m
Length of footing(L) = 4.000 m
Depth of foundation =
GL - FL = 3.000 m.

(2) Shape factor:

Shape of foundation = Square
Sc = **1.30**
Sq = **1.20**
Sr = **0.80**

(3) Depth factor:

$$dc = 1 + 0.2 \frac{x\ Df}{B} \sqrt{N\ \varnothing}$$

where,

$$\sqrt{N\ \varnothing} = \tan\left(45 + \frac{30}{2}\right)$$

$$= 1.73$$

$$dc = \mathbf{1.26}$$

$$dq = dr = 1 + 0.1 \frac{x\ Df}{B} \sqrt{N\ \varnothing}$$

For $\varnothing > 10^o$

dq = dr = **1.13**

(4) Inclination Factor:

Angle of inclination = **0.000**
ic = iq = **1.000**
ir = **1.000**

(5) Bulk Unit Wt. of Foundation Soil:

$$Y = yd\ (1 + W/100)$$

$$= \mathbf{1.640}\ t/m^3$$

(6) Effective surcharge:

$$q = Y \times Df$$

$$= yd\,(1 + w/100) \times Df$$

$$= \mathbf{2.982} \quad t/m^2$$

(7) Correction factor for Water table:

Ground level		=	10.000	m
Water table		=	10.000	m
FL - WTL	=	-3.000	<	3
FL - WTL		<	Df	
FL - WTL		**0.500**		

(8) Type of failure:

Initial void ratio

$$eo = \frac{\text{Specific gravity}}{\text{Dry density}} - 1 = \frac{2.64}{1.600} - 1$$

$$= 0.6500 > 0.55$$

$\therefore$ **Mix Shear Failure** **MSF**

(9) Factors Nc, Nq, Nr from table :1, Pg. 8 of IS:6403-1981:

Corrected $\varnothing$ = **25.72** ^{o} Corrected C = **0.000**

Nc = **21.82** Nq = **11.52** Nr = **12.08**

With the use of above parameters, safe bearing capacity is calculated as below:

<u>Net ultimate bearing capacity</u>

$$'q_d = CN_cS_cd_ci_c + q(N_q - 1)\,S_qd_qi_q + 1/2 \times B \times YNrS_rd_ri_rW'$$

$$= 0.00 + 42.51 + 10.85$$

$$qd = \mathbf{53.361} \quad t/m^2$$

<u>SAFE BEARING CAPACITY:</u>

$$SBC = \frac{\text{Net Ultimate Bearing capacity}}{\text{Factor of safety}}$$

$$SBC = \frac{53.36}{3}$$

$$SBC = 17.787 \quad t/m^2$$

$\therefore$ **Say SBC = 17.79 t/m^2**

(B) Allowable Bearing Pressure by Cosidering 50 mm Settlement.

1	Width of Foundation (B)	=	4.000		4.00	400	cm
2	Length of Foundation (L)	=	4.000		1.80	400	cm
3	Depth of Foundation (D)	=	3.000	m		300	cm
4	Field Dry Density (rd)	=	1.600		1.61	0.0016	kg/cm3
5	Field Moisture Contents (w)	=	2.650		7.90		
6	Wet Density (rb)	=	1.642		2.63	0.0016424	kg/cm3
7	Submerged Density (rsub)	=	0.992		30.00	0.00099163	kg/cm3
7	Specific Gravity (G)	=	2.630				
8	Void Ratio (eo)	=	0.644				
9	Depth of Bore hole =		10.00				
10	Depth of Water Table =		3.000				

Average N Value between base of the footing and the depth equal to 1.5 times the width of footing N avg = 16

.

Applying correction for overburden & dilatancy

(1) Overburden pressure below foundation q = r*D

= 4.9272 t/m2

q'= 4.9272 t/m2

Corection Factor =CF = 1.24

N= 16

N' = CF* N

N' = 19.82

N'' = 20

(2) Correction for Dilatancy

N'' = 15 + 0.50 (N' -15)

N'' = 17.41

N'' = 17

Taking N- Value as 17 & width of footing as 4.000 m, total settlement for 1.00 kg/cm2 pressure is 20 mm as per fig.-9 IS-8009(Part-1)

L/B	1.000
D/_/LB	0.750
_/LB/D	1.333
Depth Correction =	0.780
Rigidity Correction =	1.00
Water Table correction =	0.50
Corrected Settlement =	**31.20** mm for 1 Kg/cm2 pressure

Allowable Bearing Pressure for 50 mm Settlement = 1.603 kg/cm^2

Say Allowable Bearing Pressure = 16.03 t/m^2

7 Summary

Analysis of S.B.C.						
Depth Below existing G.L. (meter)	**Depth Below Natural G.L. (meter)**	**Type of Foundation**	**Foundation Size (mtr x mtr)**	**Net S.B.C. t/m^2**		**Recomm. Net S.B.C. t/m^2**
				Shear Criterion C & Φ Value	**Settlement Criteria (50mm)**	
2.00	2.00	Square	2.00 x 2.00	11.69	>11.00	**11.00**
			3.00 x 3.00	12.00	>11.00	**11.00**
			4.00 x 4.00	12.56	>11.00	**11.00**
2.50	2.50	Square	2.00 x 2.00	14.66	>14.00	**14.00**
			3.00 x 3.00	14.71	>14.00	**14.00**
			4.00 x 4.00	15.13	>14.00	**14.00**
3.00	3.00	Square	2.00 x 2.00	17.81	>16.00	**16.00**
			3.00 x 3.00	17.53	>16.00	**16.00**
			4.00 x 4.00	17.79	16.03	**16.00**

8 Conclusion & Recommendations:

- The study of the soil investigation at the site at various depths indicates layers as shown in the general stratifications.
- Ground water table is not encountered at the time of investigation.
- Any variation in stratification in any of the foundation location shall be studied thoroughly before executing the foundation work.
- S.B.C. is recommended by considering all the parameters affecting to soil At the field as well as in the laboratory.
- **Side soil should be retained during execution of work.**

For **Vishwas Geotech Pvt.Ltd.** Date - 28/11/2022

Sign

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NOTATIONS

C	Cohesion
ϕ	Angle of internal friction of soil
DS	Disturbed Sample
UDS	Undisturbed Sample
SPT	Standard Penetration Test
NP	Non Plastic Soils
Cc	Compression Index
Pc	Pre Consolidation pressure
TUU	Unconsolidated undrained traixial test
DUU	Unconsolidated undrained direct shear test
HWR	Highly Weathered Rock
MWR	Moderately Weathered Rock
FR	Fresh Rock
TCU	Consolidated Undrained traixial test
DCU	Consolidated Undrained Directshear test

REFERENCE

Indain Standards	IS 2720 Pt. II, III, IV, V, XIII, XXXI, XXVI, XXV IS 1498, IS 6403, IS 1904, IS 2911 IS : 12070 - 1987, IS : 8009 - Pt- 1 - 1973, IS : 4453 - 1967
Murthy V.N.S.	Soil Mechanics and Foundation engineering (Dhanpat Rai and Sons Delhi)
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Peck. R.S. Hanson W.E.	Foundation Engineering (Asia Publishing House)
Nayak N.V.	Foundation Engineering Manual (Dhanpat Rai and Sons)
Kaniraj S.R.	Design aids in soil mechnaics and foundation Engineering (Tata Mc Graw Hill Publishing Co. Ltd.)
Alam Singh	Modern Geotechnical Eng. (IBT Publishing & Distrubutors Delhi.)

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SUMMARY OF EXPLORATION DATA

TABLE - 1
Borehole - 1

Depth in mtr.	To	Sample No. & Type	Visual Description of Soil & Stratum No.	Hatching	IS Classification	Particle Size Analysis: Gravel % - 4.75 mm & above	Particle Size Analysis: Sand % - 0.075 to 4.75 mm	Particle Size Analysis: Silt % - 0.002 to 0.075 mm	Particle Size Analysis: Clay % - Less than 0.002 mm	Atterberg's Limit: Liquid Limit	Atterberg's Limit: Plastic Limit	Atterberg's Limit: Plasticity Index	Dry Density gm/cc	Natural Moisture Wn %	Specific Gravity Gs	Shear Parameters: Cohesion kg/sq.cm.	Shear Parameters: Degree	Shear Parameters: Type of Test	Consolidation Test: c.c.	Consolidation Test: p.c. kg/sq.cm.	Unconfined Comp. Strength kg/sq.cm	Srinkage Factor in %	For Rock: Water Absorption (%)	For Rock: Porosity (%)
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**		**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
0.00	1.00	DS	Non Plastic Silty Sand		SM	0	81	19		22	N P	N P		3.30										
1.50	1.95	SPT			SM	0	83	17		24	N P	N P		3.00										
3.00	3.45	UDS			SM	0	82	18		24	N P	N P	1.60	2.50	0.64	0.00	30	DUU						
4.50	4.95	SPT			SM	10	74	16		23	N P	N P		2.30										
6.00	6.45	UDS			SM	11	62	27		22	N P	N P	1.61	2.20	2.64	0.00	31	DUU						
7.50	7.95	SPT			SM	10	77	13		21	N P	N P		2.10										
9.00	9.45	UDS			SM	12	75	13		20	N P	N P	1.62	2.00	2.64	0.00	32	DUU						
10.00	10.45	SPT			SM	10	75	15		19	N P	N P		1.90										

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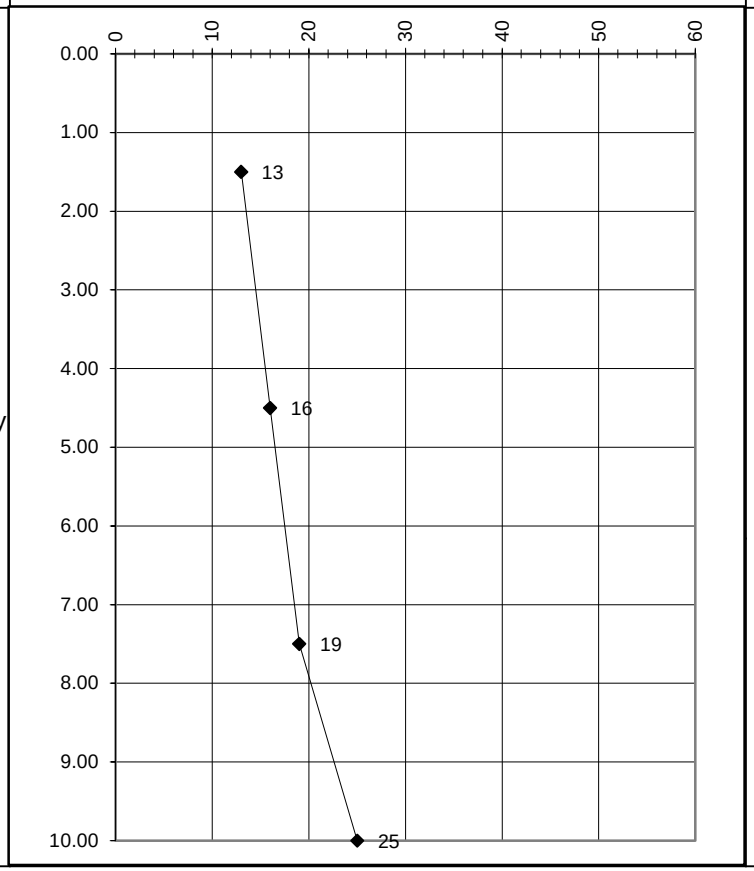
BORELOG

BORE HOLE NO : 1

Depth of Borehole : 10.45 mtr.

Method of Boring	Casing	Bore Diameter	G.W.L.	Depth (M)	Thickness of Layer (M)	Hatching	VISUAL SOIL DESCRIPTION	PENETRATION TEST (N-VALUES)	S.P.T. No. of Blows N1	N2	N3	N- Value	Un Disturbed Sample	Disturbed Sample	Remark
Hand Auger	Not Used	150 mm.	--	0.00	10.45		Non Plastic Silty Sand							0.00	
				1.00					5	6	7	13		1.50	
				2.00									3.00		
				3.00											
				4.00					6	7	9	16		4.50	
				5.00											
				6.00									6.00		
				7.00					7	9	10	19		7.50	
				8.00											
				9.00									9.00		
				10.00					10	12	13	25			



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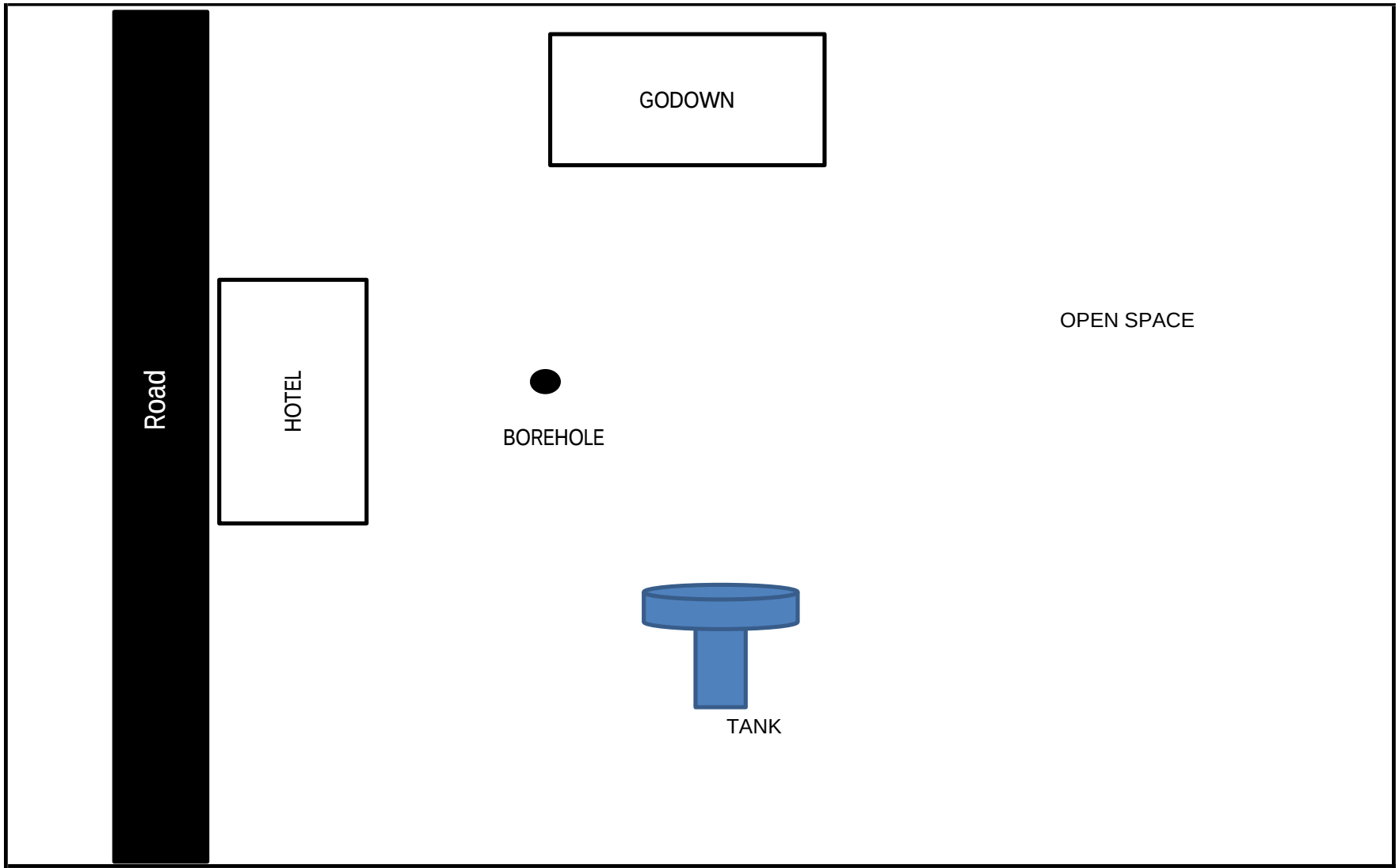

NOTE : -DRAWING IS NOT TO THE SCALE

Bore Hole Locati

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Site Photos

SITE ID - IN- 3154999	**SITE NAME.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State Gujarat, Picode 385310, Police station Samarvada**	**SITE ADD.- Open Plot of Mrs Gangaben Dathji Devda, Gamtal Plot No 227, At Moje Samarvada village, Taluka Dhanera, District Banaskantha, State Gujarat, Picode 385310, Police station Samarvada**

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